

UNIT 2 DIFFERENTIATION

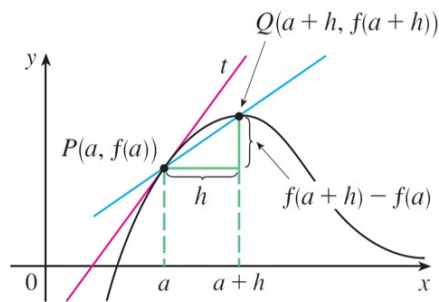
2.7 Derivatives and Rates of Change

- **Tangent Line**

The **tangent line** to the curve $y = f(x)$ at the point $P(a, f(a))$ is the line through P with slope

$$m_{\text{tan}} = \lim_{h \rightarrow 0} m_{\text{sec}} = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

provided that this limit exists and is not ∞ or $-\infty$.



- **Example** Find an equation of the tangent line to the curve $y = \frac{3}{x}$ at the point $(3, 1)$.

- **Instantaneous Velocity**

If an object moves along a coordinate line with position function $f(t)$, then its **instantaneous velocity** at time $t = a$ is

$$v = \lim_{h \rightarrow 0} v_{\text{avg}} = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

provided that the limit exists and is not ∞ or $-\infty$.

- **Derivative**

The **derivative** of a function f at a number a is

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

- If the limit exists, we say that f is **differentiable** at a . Finding a derivative is called **differentiation**.
- **Example** Given $f(x) = x^3 + 7x$, find $f'(a)$.

Exercises 2.7 7.

Exercises 2.8 13, 15, 17, 19, 21, 23.

2.8 The Derivative as a Function

- **Definition**

The derivative of a function f is another function f' whose value at any number x is

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

- **Example** Given $f(x) = \frac{1}{x}$, find $f'(x)$.

- **Example** Given $f(x) = \sqrt{x}$, find $f'(x)$.

- **Leibniz Notation for the Derivative**

$$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = f'(x)$$

- **Other Notations** For $y = f(x)$

$$f'(x) = y' = \frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx}f(x) = Df(x) = D_x f(x)$$

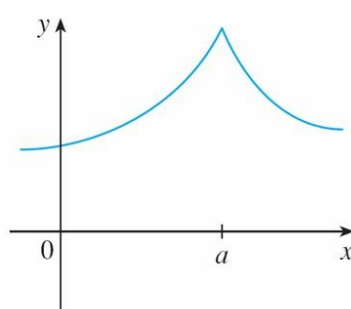
- A function is differentiable on an open interval if it is differentiable at every number in that interval.
- **Example** Where is the function $f(x) = |x|$ differentiable?

- **Differentiability Implies Continuity**

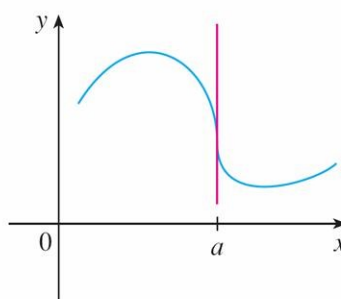
If $f'(a)$ exists, then f is continuous at a .

- Continuity does **not** imply differentiability.
- How can a function fail to be differentiable?

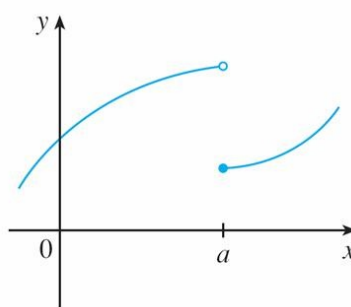
- A corner



- A vertical tangent



- A discontinuity

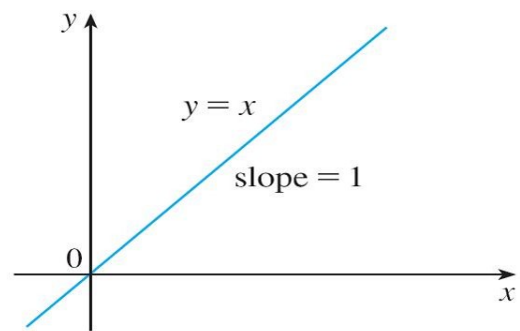
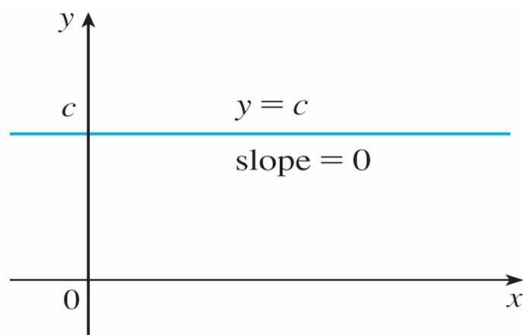


Exercises 2.9 21, 23, 25, 27, 29, 37, 43.

3.1 Derivatives of Polynomials and Exponential Functions

- Rules for Finding Derivatives

- $\frac{d}{dx}c = 0$
- $\frac{d}{dx}x = 1$
- $\frac{d}{dx}x^n = nx^{n-1}$, n is a positive integer.



- For any real number n , $\frac{d}{dx}x^n = nx^{n-1}$.

- $D_x [kf(x)] = kD_x f(x)$
- $D_x [f(x) \pm g(x)] = D_x f(x) \pm D_x g(x)$

- **Example** Find the derivatives of the following functions:

(a) $y = 5x^2 + 7x - 6$

(b) $y = 4x^6 - 3x^5 - 10x^2 + 5x + 16$

(c) $y = \frac{x^2 + 1}{x}$

(d) $y = \frac{x + 1}{\sqrt{x}}$

- **High Order Derivatives**

The derivative of f is f' . Differentiating f' will produce f'' , called the second derivative of f . In the same way, we define f''' , the third derivative of f , $f^{(4)}$, the fourth derivative of f , $f^{(5)}$, the fifth derivative of f , and so on.

- **Example** Find all the derivatives of $f(x) = 2x^3 - 6x + 8$.

- **Notations**

Derivative	f' Notation	y' Notation	D Notation	Leibniz Notation
First	$f'(x)$	y'	$D_x y$	$\frac{dy}{dx}$
Second	$f''(x)$	y''	$D_x^2 y$	$\frac{d^2 y}{dx^2}$
Third	$f'''(x)$	y'''	$D_x^3 y$	$\frac{d^3 y}{dx^3}$
Fourth	$f^{(4)}(x)$	$y^{(4)}$	$D_x^4 y$	$\frac{d^4 y}{dx^4}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
n th	$f^{(n)}(x)$	$y^{(n)}$	$D_x^n y$	$\frac{d^n y}{dx^n}$

- **Example** The equation of motion of a particle is $s = 2t^3 - 5t^2 + 3t + 4$, where s is measured in centimeters and t in seconds. Find the acceleration as a function of time. What is the acceleration after 2 seconds?

- **Derivatives of Exponential Functions**

- **The Definition of e**

e is the number such that $\lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1$.

- **Derivative of Natural Exponential Function**

$$\frac{d}{dx}e^x = e^x$$

- **Example** At what point on the curve $y = e^x$ is the tangent line parallel to the line $y = 2x$?

Exercises 3.1 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 29, 39, 45, 47.

Exercises 3.7 5, 33, 49(a).

3.2 Product and Quotient Rules

- **Example** Given $g(x) = x$, $h(x) = x^2$, and $f(x) = g(x)h(x)$, find $D_x g(x)$, $D_x h(x)$, $D_x f(x)$, and show that $D_x[g(x)h(x)] \neq [D_x g(x)][D_x h(x)]$.

- **Product Rule**

If u and v are differentiable functions of x ,

$$\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$$

- **Example** Given $f(x) = xe^x$,
(a) Find $f'(x)$.
(b) Find $f^{(n)}(x)$.
- **Example** Given $f(t) = \sqrt{t}(a + bt)$, find $f'(t)$.
- **Example** Given $f(x) = \sqrt{x}g(x)$, where $g(4) = 2$, $g'(4) = 3$, find $f'(4)$.

- **Quotient Rule**

If u and v are differentiable functions of x ,

$$\frac{d}{dx} \frac{u}{v} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

at the point where $v(x) \neq 0$.

- **Example** Find $\frac{d}{dx} \left(\frac{3x - 5}{x^2 + 7} \right)$.

- **Example** Find the equation of the tangent line to the curve $y = \frac{e^x}{1+x^2}$ at the point $\left(1, \frac{e}{2}\right)$.

Exercises 3.2 1, 3, 5, 7, 11, 17, 25, 31.

3.3 Derivatives of Trigonometric Functions

- **Special Trigonometric Limits**

$$(a) \lim_{t \rightarrow 0} \frac{\sin t}{t} = 1$$

$$(b) \lim_{t \rightarrow 0} \frac{1 - \cos t}{t} = 0.$$

- **Example** Find $\lim_{x \rightarrow 0} \frac{\sin 3x}{x}$.

- **Example** Find $\lim_{t \rightarrow 0} \frac{1 - \cos t}{\sin t}$.

- **Example** Find $\lim_{x \rightarrow 0} \frac{\sin 4x}{\tan x}$.

- **Theorem**

- $\frac{d}{dx} \sin x = \cos x$
- $\frac{d}{dx} \cos x = -\sin x$

- **Example** Find $\frac{d}{dx} (3 \sin x - 2 \cos x)$.

- **Example** Find $\frac{d}{dx} (x^2 \sin x)$.

- **Example** Find $\frac{d}{dx} \left(\frac{1 + \sin x}{\cos x} \right)$.

- **Theorem**

- $\frac{d}{dx} \tan x = \sec^2 x$
- $\frac{d}{dx} \cot x = -\csc^2 x$
- $\frac{d}{dx} \sec x = \sec x \tan x$
- $\frac{d}{dx} \csc x = -\csc x \cot x$

- **Example** Find the equation of the tangent line to the curve $y = \tan x$ at the point $\left(\frac{\pi}{4}, 1\right)$.

- **Example** Find $\frac{d}{dx}(x^n \cot x)$ where n is an integer.

Exercises 3.4 1, 3, 5, 9, 13, 15, 17, 18, 19, 20, 21, 29, 31(a), 35, 37, 43.

3.4 The Chain Rule

- **Example** Find $\frac{d}{dx}(x^2 + 2x)^3$.

- **Chain Rule**

Let $y = f(u)$ and $u = g(x)$. If g is differentiable at x and f is differentiable at $u = g(x)$, then the composite function $f \circ g$, defined by $(f \circ g)(x) = f(g(x))$, is differentiable at x and

$$(f \circ g)'(x) = f'(g(x)) g'(x)$$

or

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

- **Example** If $y = (2x^2 - 4x + 1)^{60}$, find $D_x y$.

- **Example** If $y = \frac{1}{\sqrt[3]{2x^5 - 7}}$, find $\frac{dy}{dx}$.

- **Example** Find $D_t \left(\frac{t^3 - 2t}{t^4 + 3} \right)^{13}$.

- **Example** If $y = \sin 2x$, find $\frac{dy}{dx}$.
- **Example** Find $F'(y)$ if $F(y) = y \sin(y^2)$.
- **Example** Find $D_x \frac{x^2(1-x)^3}{1+x}$.

- **Example** Find $\frac{d}{dx}e^{\sin x}$.
- **Example** Find $\frac{d}{dx}a^x$, where $a > 0$, $a \neq 1$.
- **Example** Find $\frac{d}{d\theta}2^{\sec \theta}$.

- **Example** Find $D_x \sin^3(4x)$.

- **Example** Find $D_x \sin [\cos (x^2)]$.

Exercises 3.5 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 43, 65, 79.

3.5 Implicit Differentiation

- The relation $y^3 + 7y = x^3$ can define y as a function of x . It is called an **implicit function**.
- **Example** Find $\frac{dy}{dx}$ if $y^3 + 7y = x^3$.
- **Example** Given $\sin(x + y) = y^2 \cos x$, find $\frac{dy}{dx}$.

- **Example** Find the equation of the tangent line to the curve $4x^2y - 3y = x^3$ at the point $(1, 1)$.

- **Derivatives of Inverse Trigonometric Functions**

- $D_x \sin^{-1} x = \frac{1}{\sqrt{1-x^2}}$

- $D_x \cos^{-1} x = -\frac{1}{\sqrt{1-x^2}}$

- $D_x \tan^{-1} x = \frac{1}{1+x^2}$

- **Example** Differentiate $y = \frac{1}{\sin^{-1} x}$.

- **Example** Differentiate $y = x \arctan \sqrt{x}$.

Exercises 3.6 5, 7, 9, 11, 13, 15, 17, 25, 41, 43, 53.

3.6 Derivatives of Logarithmic Functions

- Derivatives of Logarithmic Functions

- $D_x \ln x = \frac{1}{x}$

- $D_x \log_a x = \frac{1}{x \ln a}$

- **Example** Find $D_x \ln \sqrt{x}$.

- **Example** Find $D_x \ln(x^2 - x - 2)$.

- **Example** Show that

$$D_x \ln |x| = \frac{1}{x}$$

- **Example** Find $\frac{d}{dx} \ln(1 + \cos x)$.
- **Example** Find $\frac{d}{dx} \sqrt{\ln x}$.
- **Example** Find $\frac{dy}{dx}$ if $y = \ln \sqrt[3]{\frac{x-1}{x^2}}$, $x > 1$.
- **Example** Differentiate $y = \frac{\sqrt{1-x^2}}{(x+2)^{\frac{2}{3}}}$.

- **Example** Differentiate $y = x^x$.

- **Example** Differentiate $y = (x^2 + 1)^{\sin x}$.

- **e as a limit**

$$\lim_{x \rightarrow 0} (1 + x)^{\frac{1}{x}} = e$$
$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$$

- **Example** Find $\lim_{n \rightarrow \infty} \left(\frac{n+2}{n}\right)^n$.

- **Example** Find $\lim_{n \rightarrow \infty} \left(\frac{n-2}{n}\right)^n$.

Exercises 3.8 3, 5, 7, 9, 11, 15, 19, 35, 39, 41, 43.

10.1 Parametric Equations

- A plane curve can be defined by a pair of **parametric equations**

$$x = f(t), \quad y = g(t), \quad t \in [a, b]$$

- $P(f(a), g(a))$ is called the **initial point**.
- $Q(f(b), g(b))$ is called the **terminal point**.
- **Example** Sketch and identify the curve defined by the parametric equations

$$x = t^2 - 2t, \quad y = t + 1, \quad -2 \leq t \leq 4$$

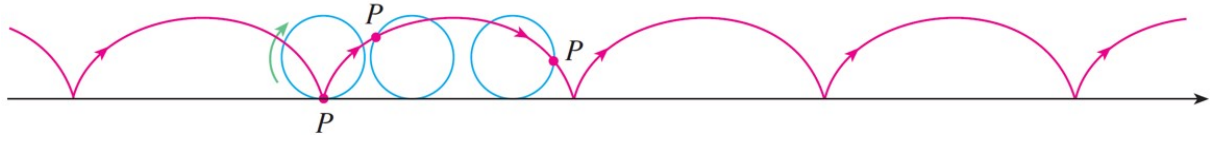
t	x	y
-2		
-1		
0		
1		
2		
3		
4		

- **Example** Eliminate the parameter in

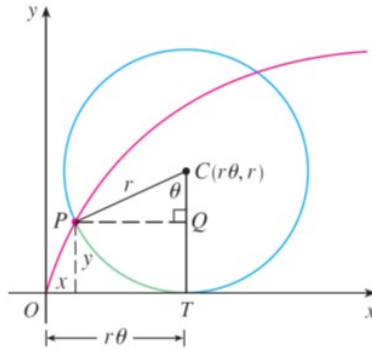
$$x = \cos t, \quad y = \sin t, \quad 0 \leq t \leq 2\pi$$

and determine the curve it represents.

- **Example** A cycloid is the curve traced by a point P on the rim of a wheel as the wheel rolls along a straight line without slipping.



Assume that the radius of the wheel is r . Derive parametric equations for the cycloid as follows. The parameter is θ .



- (a) Show that $|OT| = r\theta$.
- (b) Write $|PQ|$ and $|QC|$ in terms of r and θ .
- (c) Show that $x = r(\theta - \sin \theta)$ and $y = r(1 - \cos \theta)$.

Exercises 10.1 5, 7, 9, 11, 13, 39.

10.2 Calculus with Parametric Curves

- **Calculus for Curves Defined Parametrically**

Let f and g be continuously differentiable with $f'(t) \neq 0$ on $\alpha < t < \beta$. Then the parametric equations

$$x = f(t), \quad y = g(t)$$

define y as a differentiable function of x and

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

Horizontal tangents:

Vertical tangents:

- **Example** A curve C is defined by the parametric equations $\begin{cases} x = t^2 \\ y = t^3 - 3t \end{cases}$.
 - (a) Show that C has two tangents at the point $(3, 0)$ and find their equations.

(b) Find the points on C where the tangent is horizontal or vertical.

- **Example** Recall that the cycloid has parametric equations $\begin{cases} x = r(t - \sin t) \\ y = r(1 - \cos t) \end{cases}$.
 - (a) Find the equation of the tangent line to the cycloid at the point $t = \frac{\pi}{3}$.

- (b) At what point is the tangent horizontal?

Exercises 10.2 3, 17, 19, 25.

4.1 Maxima and Minima

- **Definition**

Let D , the domain of f , contain the point c . We say that

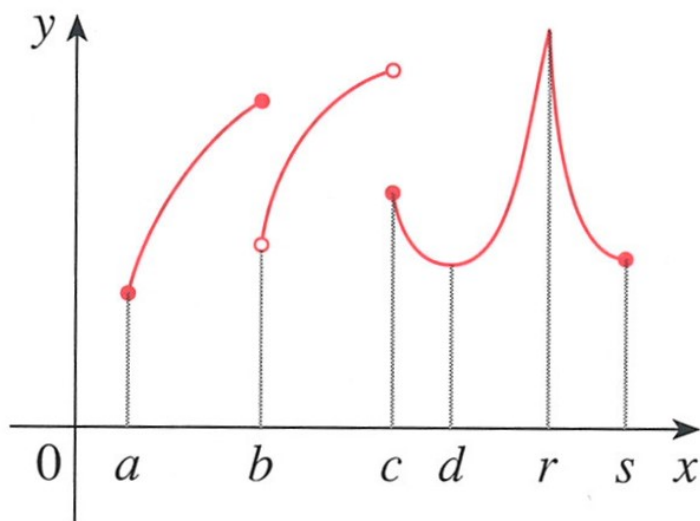
- (i) $f(c)$ is the **absolute maximum** value of f on D if $f(c) \geq f(x)$ for all x in D ;
- (ii) $f(c)$ is the **absolute minimum** value of f on D if $f(c) \leq f(x)$ for all x in D ;
- (iii) $f(c)$ is an absolute extreme value of f on D if it is either the absolute maximum value or the absolute minimum value;

- **Definition**

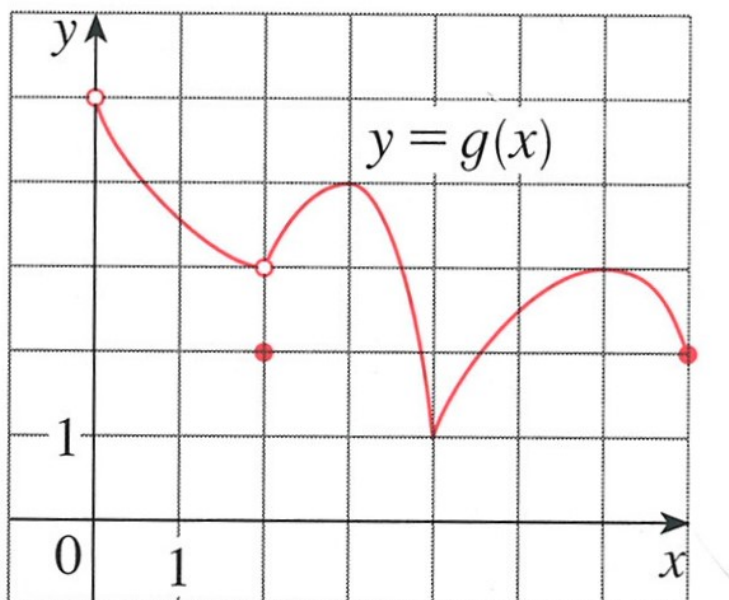
Let D , the domain of f , contain the point c . We say that

- (i) $f(c)$ is a **local maximum** value of f if $f(c) \geq f(x)$ when x is near c ;
- (ii) $f(c)$ is a **local minimum** value of f if $f(c) \leq f(x)$ when x is near c ;
- (iii) $f(c)$ is a local extreme value of f on D if it is either a local maximum value or a local minimum value.

- **Example** Find the absolute extreme values and the local extreme values of the function.



- **Example** Find the absolute extreme values and the local extreme values of the function.



- **Extreme Value Theorem**

If f is continuous on a closed interval $[a, b]$, then f attains both an absolute maximum value and an absolute minimum value there.

- If c is a point at which $f'(c) = 0$, we call c a **stationary point**.
- **Example** Find the stationary points of the function $f(x) = 2x^3 - 3x^2 + 2$.
- If c is an interior point of the domain of f such that $f'(c)$ does not exist, we call c a **singular point**.
- **Example** Find the singular points of the function $y = \sqrt[3]{x^2}$.

- If f is a function defined on an interval, a **critical point** c of f is one of the following three types of points:
 - (a) an end point of I ;
 - (b) a stationary point of f ;
 - (c) a singular point of f .

- **Fermat's Theorem**

Let f be defined on an interval I containing the point c . If $f(c)$ is a local extreme value, and $f'(c)$ exists, then $f'(c) = 0$.

- **Critical Point Theorem**

Let f be defined on an interval I containing the point c . If $f(c)$ is an extreme value, then c must be a critical point.

- In order to find the extreme values of a function f that is continuous on a closed interval $I = [a, b]$,
 - Step 1 Find the critical points of f on I .
 - Step 2 Evaluate f at each of these critical points. The largest value is the maximum value; the smallest value is the minimum value.

- **Example** Find the absolute maximum and absolute minimum values of the function $f(x) = x^3 - 3x^2 + 1$ on $\left[-\frac{1}{2}, 4\right]$.

- **Example** Find the maximum and minimum values of $F(x) = x^{\frac{2}{3}}$ on $[-1, 2]$.

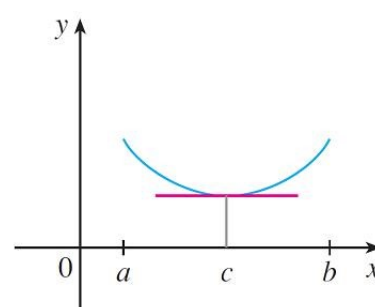
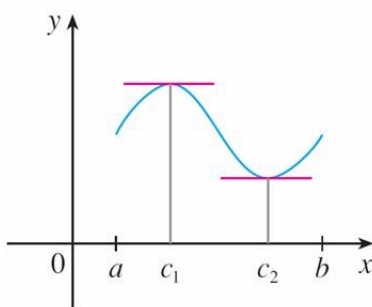
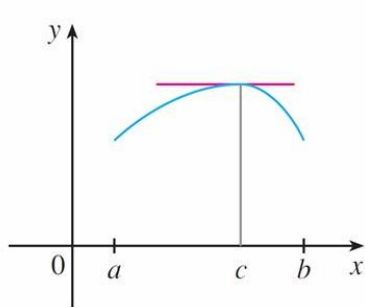
Exercises 4.1 3, 5, 29, 47, 53, 57.

4.2 The Mean Value Theorem for Derivatives

• Rolle's Theorem

If f is continuous on a closed interval $[a, b]$ and differentiable on its interior (a, b) , and $f(a) = f(b)$, then there is a c in (a, b) where

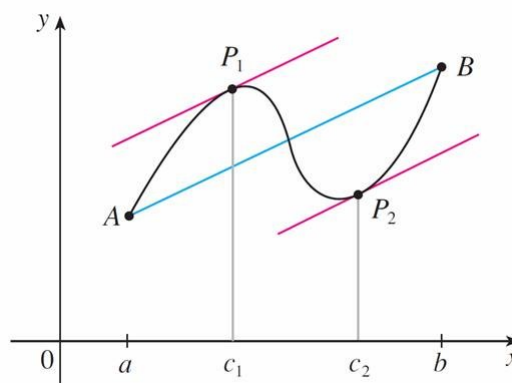
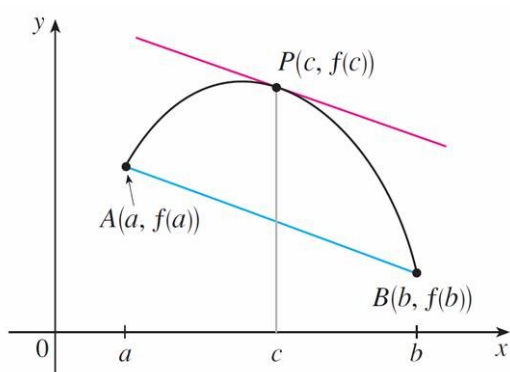
$$f'(c) = 0$$



• The Mean Value Theorem for Derivatives

If f is continuous on a closed interval $[a, b]$ and differentiable on its interior (a, b) , then there is a c in (a, b) where

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$



- **Example** Prove that the equation $x^3 + x - 1 = 0$ has exactly one real root using the Intermediate Value Theorem and the Mean Value Theorem.

- **Example** If $f(0) = -3$ and $f'(x) \leq 5$ for all x , how large can $f(2)$ possibly be?

- **Theorem**

If $F'(x) = 0$ for all x in (a, b) , then $F(x)$ is a constant C on (a, b) .

If $F'(x) = G'(x)$ for all x in (a, b) , then there is a constant C such that

$$F(x) = G(x) + C$$

for all x in (a, b) .

4.3 Monotonicity and Concavity

- **Monotonicity Theorem**

Let f be continuous on an interval I and differentiable at every interior point of I .

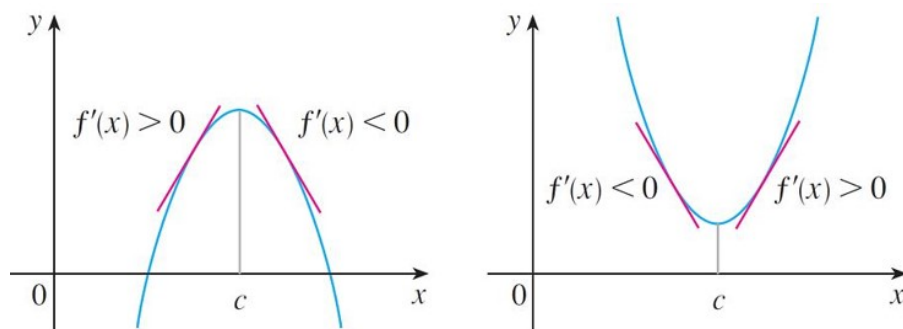
- (i) If $f'(x) > 0$ for all x interior to I , then f is **increasing** on I .
- (ii) If $f'(x) < 0$ for all x interior to I , then f is **decreasing** on I .

- **Example** If $f(x) = 2x^3 - 3x^2 - 12x + 7$, find where f is increasing and where it is decreasing.

• Local Extreme Values – First Derivative Test

Let f be continuous on an open interval (a, b) that contains a critical point c .

- (i) If $f'(x) > 0$ for all x in (a, c) and $f'(x) < 0$ for all x in (c, b) , then $f(c)$ is a **local maximum** value of f .
- (ii) If $f'(x) < 0$ for all x in (a, c) and $f'(x) > 0$ for all x in (c, b) , then $f(c)$ is a **local minimum** value of f .
- (iii) If $f'(x)$ has the same sign on both sides of c , then $f(c)$ is not a local extreme value of f .

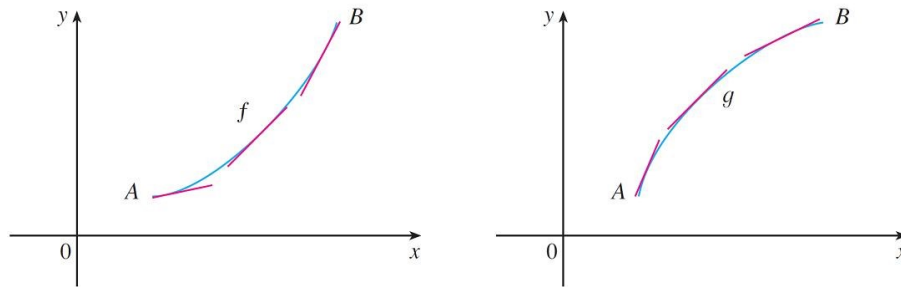


- **Example** Find the local extreme values of $f(x) = x^3 - 3x^2 - 9x + 12$ on $(-\infty, \infty)$.

- **Example** Find the local extreme values of $f(x) = x + 2 \sin x$ on $0 \leq x \leq 2\pi$.

- **Definition**

If the graph of f lies above all its tangent lines on an interval I , then we say that f is **concave upward** on I . If the graph of f lies below all its tangent lines on an interval I , then we say that f is **concave downward** on I .



Let f be differentiable on an open interval I . We say that f is **concave up** on I if f' is increasing on I , and we say that f is **concave down** on I if f' is decreasing on I .

- **Concavity Theorem**

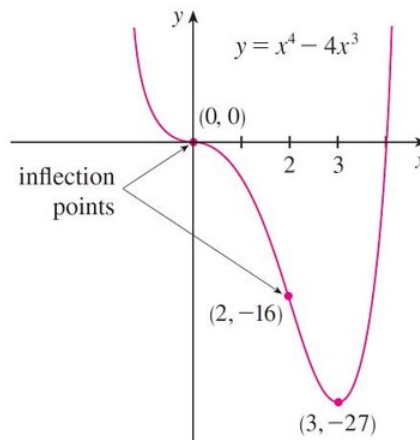
Let f be twice differentiable on an open interval I .

- (a) If $f''(x) > 0$ for all x in I , then f is **concave up** on I .
- (b) If $f''(x) < 0$ for all x in I , then f is **concave down** on I .

- **Example** Where is $f(x) = x^3 - 3x^2 - 9x + 12$ concave up and concave down?

- **Inflection Points**

We call $(c, f(c))$ an **inflection point** of the graph f if f is concave up on one side of c and concave down on the other side.



- If $(c, f(c))$ is an inflection point, then either $f''(c) = 0$ or $f''(c)$ does not exist.
- **Example** Find all the inflection points of $F(x) = x^{\frac{1}{3}} + 2$.

- **Local Extreme Values – Second Derivative Test**

Let f' and f'' exist at every point in an open interval (a, b) containing c , and suppose $f'(c) = 0$.

- (i) If $f''(c) < 0$, then $f(c)$ is a **local maximum** value of f .
- (ii) If $f''(c) > 0$, then $f(c)$ is a **local minimum** value of f .

- **Example** Find (if any) the minimum and maximum values of $f(x) = x^4 - 4x$ on $(-\infty, \infty)$.

Exercises 4.3 11, 13, 15, 17, 19, 21.

4.4 Indeterminate Forms and L'Hôpital's Rule

- **L'Hôpital's Rule**

Suppose

$$\lim_{x \rightarrow a} f(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = 0$$

or

$$\lim_{x \rightarrow a} f(x) = \pm\infty \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = \pm\infty$$

then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

if the limit on the right hand side exists.

- **Example** Use L'Hôpital's rule to evaluate $\lim_{x \rightarrow 2} \frac{x-2}{x^2-4}$ and $\lim_{x \rightarrow 0} \frac{\sin x}{x}$.

- **Example** Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$.

- **Example** Evaluate $\lim_{x \rightarrow 1} \frac{\ln x}{x - 1}$.

- **Example** Evaluate $\lim_{x \rightarrow 0} \frac{\tan 2x}{\ln(1 + x)}$.

- **Example** Evaluate $\lim_{x \rightarrow 0} \frac{\sin x - x}{x^3}$.

- **Example** Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2 + 3x}$.
- **Example** Show that for any positive real number a , $\lim_{x \rightarrow \infty} \frac{x^a}{e^x} = 0$.
- **Example** Show that for any positive real number a , $\lim_{x \rightarrow \infty} \frac{\ln x}{x^a} = 0$.

Exercises 4.4 5, 7, 9, 11, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 37 (*hint*: rewrite it as $\frac{\ln x}{\frac{1}{\sqrt{x}}}$), 41 (*hint*: rewrite it as $\frac{x^3}{e^{x^2}}$).

4.7 Optimization Problems

- **Example** A rectangular box is made from a piece of cardboard 24 cm long and 9 cm wide by cutting out identical squares from the four corners and turning up the sides. Find the dimensions of the box of maximum volume. What is the maximum volume?

- **Example** A farmer has 100 meters of wire fence with which he plans to build two identical adjacent pens. What are the dimensions of the enclosure that has maximum area?

- **Example** Find the dimensions of the largest rectangle that can be inscribed in a semicircle of radius r .

Exercises 4.7 5, 9, 11, 19.

4.9 Antiderivatives

- **Definition**

We call F an **antiderivative** of f on the interval I if $D_x F(x) = f(x)$ on I , that is, if $F'(x) = f(x)$ for all x in I .

- **Example** Find an antiderivative of the function $f(x) = 4x^3$ on $(-\infty, \infty)$.

If $F(x)$ is an antiderivative of $f(x)$, so is $F(x) + C$ for any constant C .

- **Example** Find the antiderivative of $f(x) = x^2$ on $(-\infty, \infty)$.

- **Example** Find all the functions g so that

$$g'(x) = 4 \sin x + \frac{2x^5 - \sqrt{x}}{x}$$

- **Example** Find f if $f'(x) = e^x + \frac{20}{1+x^2}$ and $f(0) = -2$.

Exercises 4.10 1, 5, 9, 11, 13, 25, 27, 29, 31.